Unit 1 Foundation of database concepts

Work through this revision selection before checking the separate answer book. The full lesson and exercise bank is available on StudyLab.

SQL schema: Students(StudentID, StudentName, City, Age); Courses(CourseID, CourseName, Fee); Enrolments(EnrolmentID, StudentID, CourseID, Mark).

1.1 Database components and models

A college stores comma-separated course names in a student spreadsheet. Design three tables and identify two DBMS services it needs.

Your answer

1.2 Keys, tables and design quality

Explain why StudentName is a weak primary key and choose a better key. Give one insertion and one update anomaly in a single enrolment spreadsheet.

Your answer

1.3 SQL Server system databases

Match a scheduled job, a temporary table, a new-database template and instance login metadata to the appropriate system database.

Your answer

Native T SQL tasks use SQL Server and SSMS in a disposable practice database. Browser tasks use the documented SQLite compatibility subset.

Unit 2 Tools of the trade

Revision selection Original supplementary material by Lefa Malatji

2.1 Tools, SELECT and filtering

Find students aged between 20 and 24 inclusive whose names begin with L. Sort oldest first, breaking ties by StudentID.

Your answer

2.2 Aggregates, DISTINCT and grouping

Show cities with at least five students aged 20 or more. Explain why the age condition belongs in WHERE.

Your answer

2.3 Safe data changes and system functions

Write an INSERT for student 99, then a scoped UPDATE and DELETE. Explain what happens if WHERE is omitted.

Your answer

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Unit 3 JOINs

Revision selection Original supplementary material by Lefa Malatji

3.1 INNER JOIN and relationship cardinality

Join Students, Enrolments and Courses to show names, course names and marks of 75 or more.

Your answer

3.2 Outer joins, CROSS JOIN and self joins

How many rows does CROSS JOIN create for 20 students and 4 courses? Write a self join listing same-city student pairs once each.

Your answer

3.3 UNION and UNION ALL

Combine the student IDs from course 1 and course 2, first retaining duplicates and then removing them. Explain the difference.

Your answer

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Unit 4 Creating and altering tables

Revision selection Original supplementary material by Lefa Malatji

4.1 Schemas, databases and data types

Choose types for a learner name, birth date, price and telephone number. Explain why a phone number should not normally be INT.

Your answer

4.2 ALTER, DROP and change planning

Write a safe plan to make an existing Email column NOT NULL. What must happen before ALTER COLUMN?

Your answer

4.3 SSMS design view and generated scripts

Create an Assessments table in design view and generate its CREATE script. State five things to compare against your model.

Your answer

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Unit 5 Constraints and referential integrity

Revision selection Original supplementary material by Lefa Malatji

5.1 Entity integrity and named constraints

Explain why CHECK(Mark BETWEEN 0 AND 100) alone does not require a mark and why a default does not forbid other values.

Your answer

5.2 Foreign keys and referential actions

Choose referential actions for financial invoices and optional staff assignment. Explain both choices.

Your answer

5.3 Add, remove and review constraints

A table contains negative prices. Can a validated CHECK(Price>=0) simply be added? Give the repair sequence.

Your answer

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Unit 6 Normalisation and other basic design issues

Revision selection Original supplementary material by Lefa Malatji

6.1 Subqueries, EXISTS and OUTPUT

Write native SQL Server OUTPUT showing old and new ages for a change to student 1.

Your answer

6.2 Normalisation with dependency reasoning

Normalise Enrolment(StudentID,CourseID,StudentName,CourseName,LecturerID,LecturerName,Mark) to 3NF. The candidate key is (StudentID,CourseID). StudentID determines StudentName; CourseID determines CourseName and LecturerID; LecturerID determines LecturerName; the full key determines Mark. Assume one lecturer per course.

Your answer

6.3 ERDs, object properties and denormalisation

A dashboard is slow. Should you immediately copy course names and totals into every student row? Propose an evidence-based alternative.

Your answer

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Unit 7 SQL Server storage, index structures and views

Revision selection Original supplementary material by Lefa Malatji

7.1 Storage hierarchy, scans and seeks

Compare a clustered index, nonclustered index and heap. Why might a scan be cheaper than a seek?

Your answer

7.2 Index creation, order and maintenance

Propose an index for enrolment lookup by CourseID ordered by Mark descending, returning StudentID. State its cost.

Your answer

7.3 Views and update restrictions

Why can a grouped view returning City and COUNT(*) not simply accept an inserted student?

Your answer

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Unit 8 Writing scripts and batches

Revision selection Original supplementary material by Lefa Malatji

8.1 Scripts, batches and variable scope

In SSMS, run DECLARE @n INT=2; then GO on its own line, then SELECT @n; on the next line. Explain the error, correct the script and describe how an application should handle GO.

Your answer

8.2 IF, WHILE and CASE

Write a CASE returning Distinction for marks >=75, Pass for >=50, Pending for NULL and Fail otherwise. Why is order important?

Your answer

8.3 System functions and TRY CATCH

Compare @@ERROR and TRY/CATCH. Explain why @@IDENTITY may not be the intended newly inserted ID.

Your answer

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Unit 9 Stored procedures and error handling

Revision selection Original supplementary material by Lefa Malatji

9.1 Create, execute, alter and drop procedures

The procedure dbo.usp_StudyLabStudentsByCity accepts @City VARCHAR(60) and selects StudentID and StudentName from dbo.Students for that city. Explain how to change it and remove it. Give two advantages of stored procedures and one maintenance cost.

Your answer

9.2 Input, OUTPUT, defaults and RETURN

dbo.usp_StudyLabCountStudents has parameters @City VARCHAR(60)='Polokwane' and @Total INT OUTPUT. It assigns the city student count to @Total and returns status 0. Call it for Pretoria, capture the count and status, and explain what breaks if the caller omits OUTPUT.

Your answer

9.3 Procedure validation and transaction errors

dbo.usp_StudyLabChangeAge accepts @StudentID INT and @Age INT. It rejects NULL or ages below 18 and unknown students, owns its transaction, and rejects calls inside another transaction. Design three tests and explain why parameter validation alone is insufficient for referential integrity.

Your answer

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